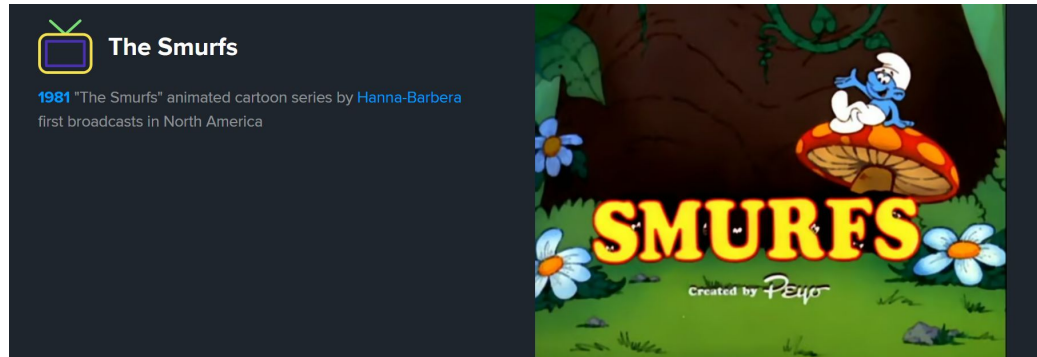


COMS W4156 Advanced Software Engineering (ASE)

September 12, 2024

On this day ...



Agenda

1. What is a Service and How is it Different from a Library?
2. Clean Code demo: Vrinda Ahuja
3. Upcoming



You Have All Used Standard Libraries

```
#include <stdio.h>

int main() {
    printf("Hello World\n");
    return 0;
}
```

```
#include <iostream>
int main() {
    std::cout << "Hello World!";
    return 0;
}
```

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

- C standard library, or libc, provides stdio (printf)
- C++ standard library provides iostream (cout)
- Java standard library provides java.lang (String and System)

You May Have Used Third-Party Libraries in your Code

OpenSSL

TensorFlow

SQLite

log5j

Apache Commons

First party = software developed by you as developer, your team, your organization.
(Standard libraries usually trusted as if first party, but you do not change!)

Second party = your clients, customers and/or users

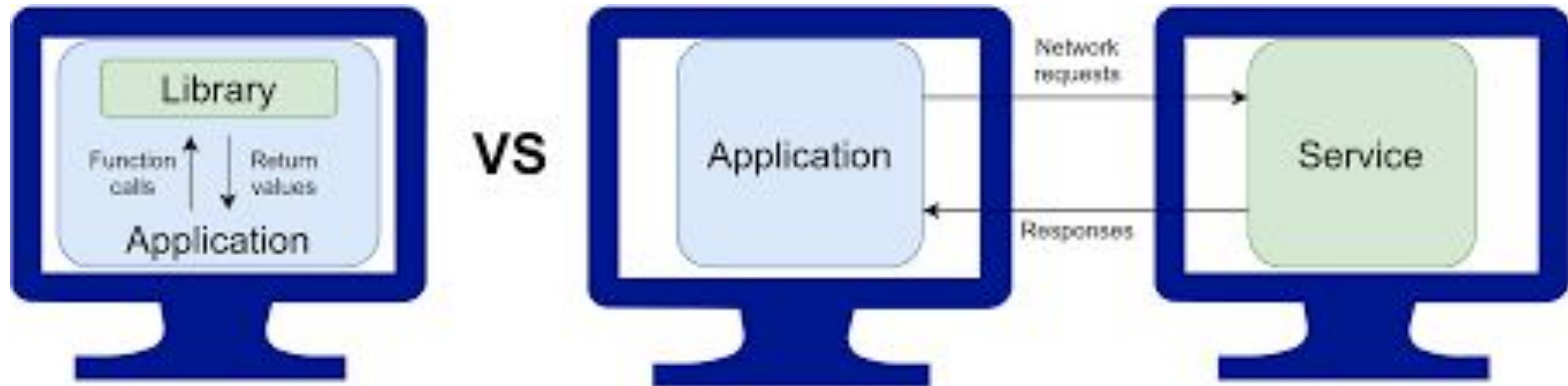
Third party = dependency developed and/or controlled by some external supplier, vendor or organization (possibly untrusted: important for security concerns later in course)



Libraries Run On The Host Machine

When you are running software on your machine, then the libraries used in your software's code are compiled or bundled together with your software and also execute on your machine

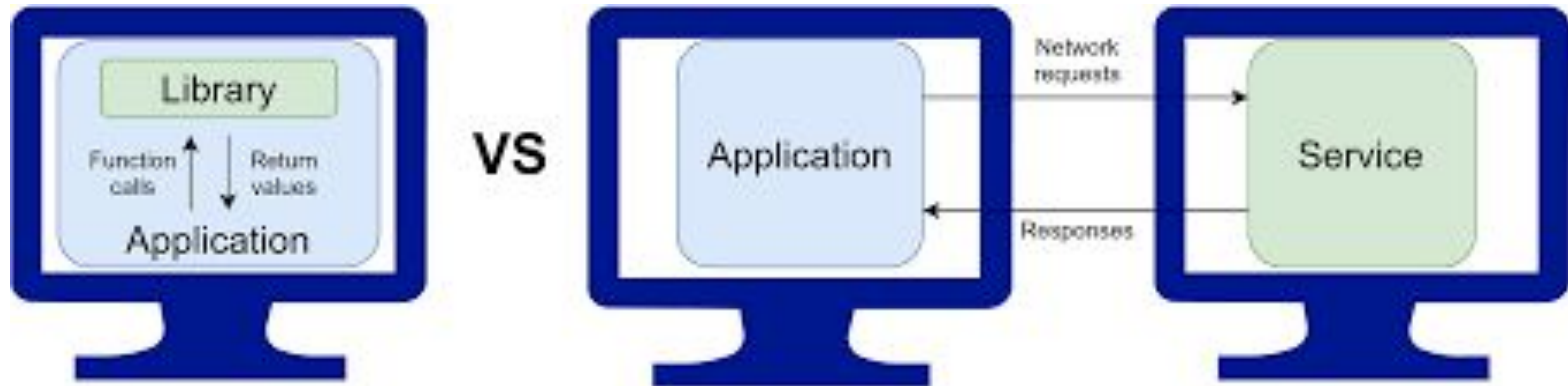
Your software has its own copy of the library (since this is an SE course, not an OS course, we do not discuss [shared libraries, memory mapping](#), etc.)



Services Run Somewhere On The Network

When you are running software on your machine, then the services used by your software's code can execute anywhere reachable from your machine (some services provide [client libraries](#) to do the network or interprocess communication)

Your software might share the service with other instances of your software and with independent software from other developers, teams, organizations



You May Have Used [Third-Party Services](#) in your Code

[PayPal Complete Payments](#)

[Google Maps Platform](#)

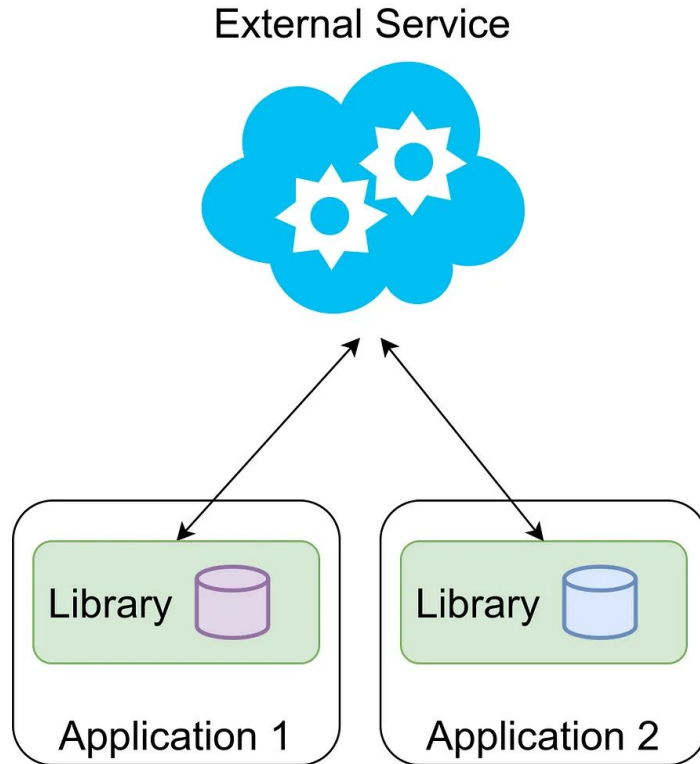
[Zoom Video SDK](#)

[OpenAI Platform](#)

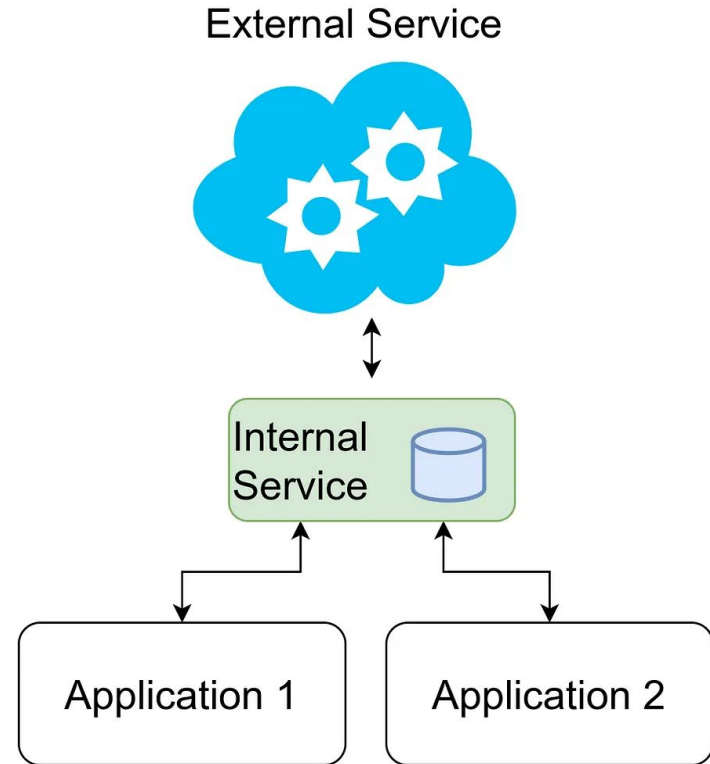
[Duo Auth API](#)



Libraries and Services Can Themselves Use Services

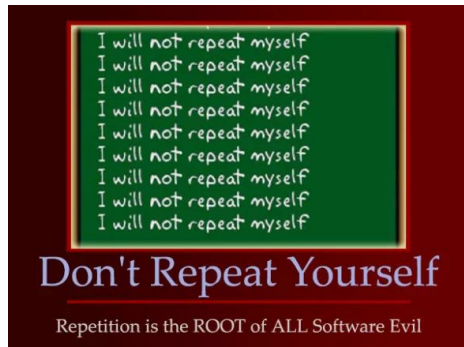


VS



Libraries and Services Can Be First Party

If you as developer, your team, your organization has a large long-lived codebase, there is probably some [duplicate code](#)



Not necessarily copy/paste "clones" (discussed further later in course), there could also be multiple independent implementations of essentially the same functionality

This violates the [DRY software development principle](#) = Don't Repeat Yourself

Duplicated code usually means more code that needs to be maintained

- If you need to fix a bug, you need to fix it in all duplicated places
- If you need to enhance the features, you need to do it in multiple parts of the code

Instead, the duplicated software should be packaged into a library or service shared across the codebase. In-house services often called "microservices"

How To Decide Between Developing Library vs (Micro)Service

Speed of Calls - nanoseconds vs milliseconds (for communication, not actual work)

Reliability of Calls - services subject to network failures and timeouts

Infrastructure - service often has its own logging, monitoring, alerting, and potentially even an on-call tech support team

Development Overhead - [secrets management](#) and authentication/authorization in addition to above infrastructure, so only authorized users/code can access service



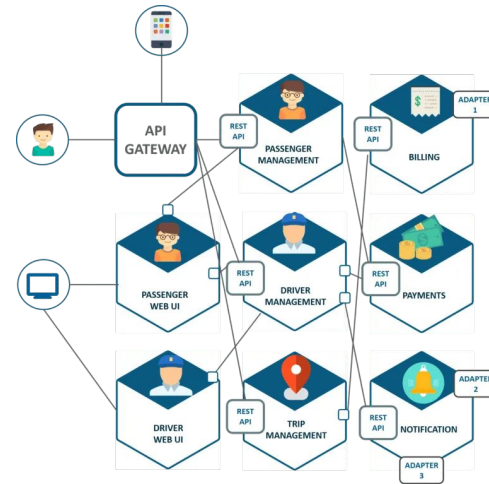
Flexibility in Development - the same service can often be easily accessed from software written in multiple languages, whereas library [foreign function interfaces](#) are more complicated

Shared State - (typically but not always) libraries stateless, services stateful

How To Decide 2

Scalability - scaling vertically (bigger machine, scaling up) vs scaling horizontally (multiple machines, scaling out)

Example: [Uber Scaling from Monolithic to Microservices](#)



Libraries and Services Are Normally Accessed via APIs

Hello World uses a standard library by calling functions exported by the library

The set of exported functions is the library's API (Application Programming Interface)

Most library use is by direct synchronous calls to API functions (simple call/return)

Service API functions ("endpoints") might also be called synchronously, but not so simple



Blackbox vs. Whitebox APIs

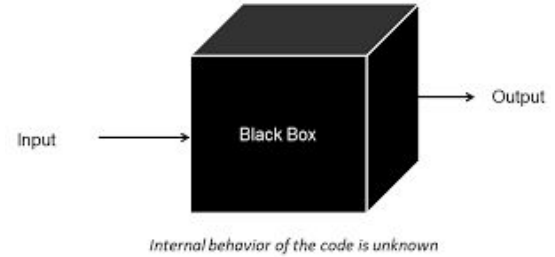
When an in-house library or service is culled from existing code in your codebase, it's probably

"whitebox" = there's nothing stopping you from reading the library/service code

Whereas third-party libraries are typically "blackbox" = available only in binary or obfuscated form, although you could potentially [decompile](#) or deobfuscate

And third-party services are almost always "blackbox" = only the API is public

Ideally, all APIs are treated as blackbox: your code uses the interface provided by the box but you should not (or can not) peek inside the box to see how it works. Your code should not depend on how it works internally because this can change without changing the interface



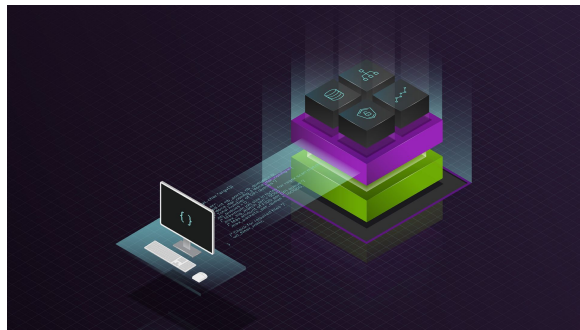
Advanced APIs

Some library APIs support

- [asynchronous messages](#) (a thread running your code sends a message to a thread running the library's code but does not block on response)
- [callbacks](#) (you register functions in your code for the library code to call)

Many service APIs support

- [asynchronous messages](#) (your code sends a message to a service but does not block on response, may block later when it "needs" the response)
- [callbacks](#) (you register functions in your code for the service to call, possibly in response to your previous asynchronous messages)

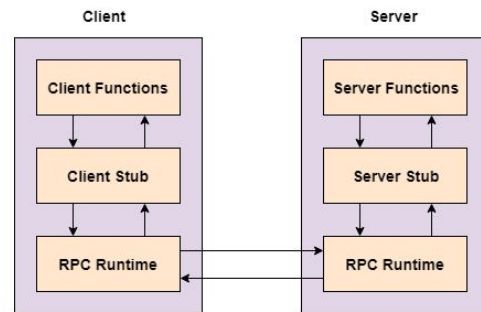
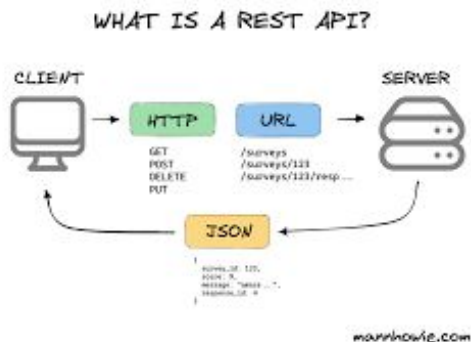


Calls to Library functions vs. Calls to Service functions

Your code calls a library function just like any other function

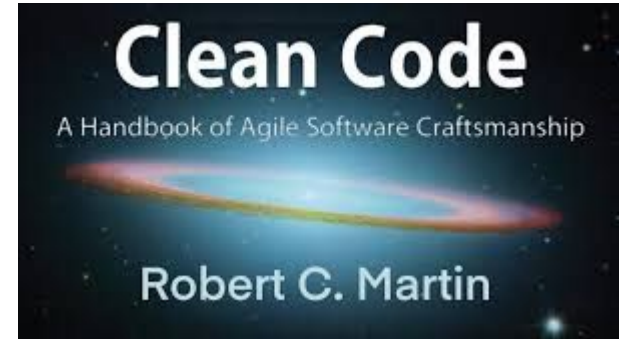
Your code requires additional infrastructure to call a service function (more on this next week). Examples:

- REpresentational State Transfer ([REST](#)), e.g., [Spring Boot](#)
- Remote procedure call ([RPC](#)), e.g., [gRPC](#)



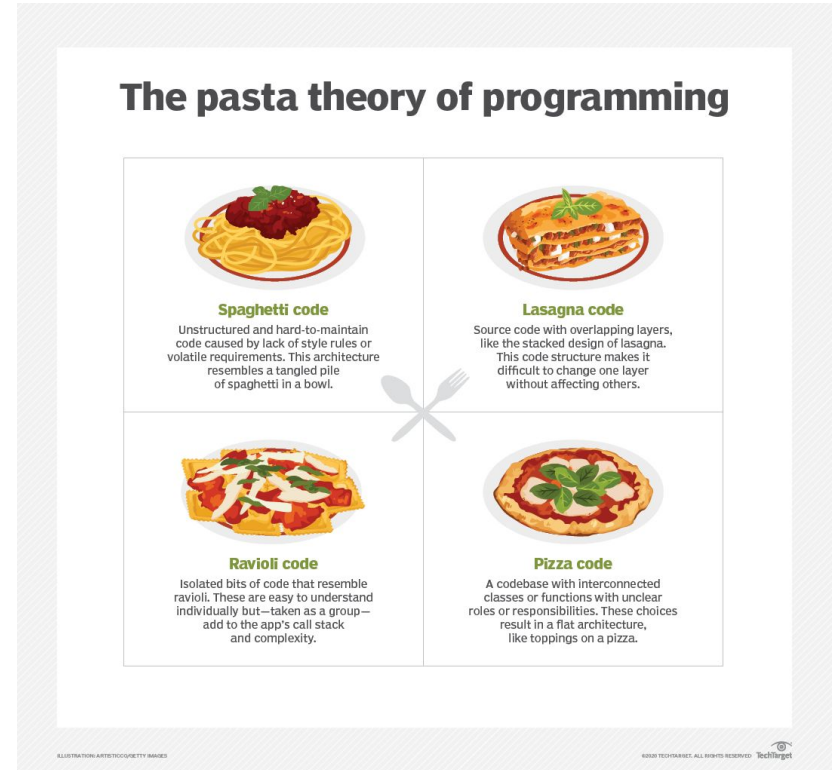
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Vrinda Ahuja Clean Code Demo

[demo slides](#)



Also see GNU humor "[Spaghetti Code](#)"

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Deadline Reminders

[Individual Miniproject part 1](#) due tomorrow September 13 11:59pm

➔ If you don't think you can submit part 1 on time, please drop the class (or drop off the waitlist) TODAY to let me add someone else

[Part 2](#) due September 20

[Part 3](#) due September 27

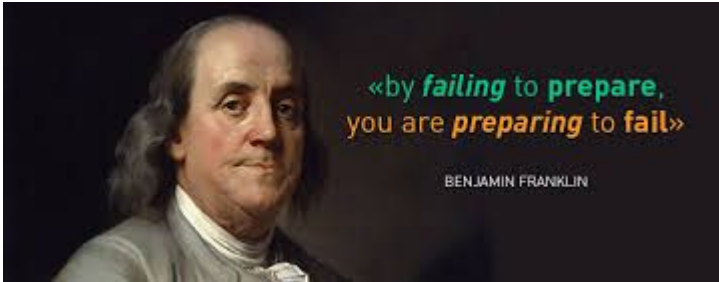


Planning Ahead to Team Project

Team Formation due September 20

Team Project Proposal due
September 27

(Note these deadlines coincide with
Miniproject deadlines, do not wait!)



Ideally 4 students per team, 5 students ok

No teams of 1, 2, 3, 6, 7, ...

Sanity Checks:

- Everyone fluent in team's choice of Java or C++
- Everyone has same development platform - Linux, Mac or Windows
- Everyone has compatible schedules
- ***Everyone is enrolled in this course and did not drop***

Next Week



Finish anything we didn't get to today

More on APIs

RESTful API demos (Java & C++)

Continuous Integration with demo

Attendance/participation requirement
starts next week...

Reminder: Class Participation (10%)

10% matters = difference between A and B!

Half (5%) based on attendance at Zoom class sessions after program change period ends. We will use Zoom's built-in reports to determine who attended and for how long (at least an hour, unless class ends early).

So make sure your Zoom name is the same as your courseworks name. If you are disconnected and reconnect, use the same name!

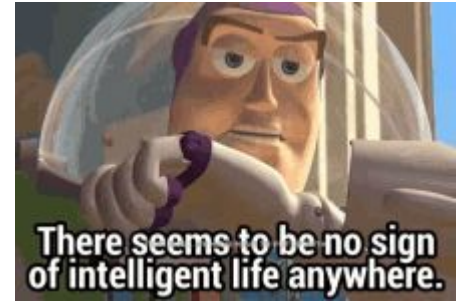


CVN students who cannot attend Zoom class sessions should do the "[CVN-only](#)" assignment for each session they miss

Non-CVN students can do the CVN-only assignment **only** if they have an acceptable (to me) excuse. Example acceptable: religious observance, illness or injury *sufficiently severe* that you cannot attend on Zoom, job interview. Example unacceptable: Forgot, slept late, busy

Reminder: Class Participation (10%) part 2

Other half (5%) based on saying something non-trivial during class and/or posting something non-trivial on [edstem](#) *throughout semester*, not just once



Example trivial: "yes", "no", "thank you", "me too", "when is xxx due?", "can I do something that you already said is not allowed?", etc.

During Zoom class sessions, you can talk using your microphone and/or you can type public messages into Zoom chat. Zoom records both, make sure to use your courseworks name!

Outside class, you can post something of likely interest to other students in the class. Must be **public** post that other students can see and comment on. Again, use your courseworks name! CVN students should post *in addition to* their [CVN-only](#) assignments (which only count for attendance)



Ask Me Anything